



ADDIS ABABA UNIVERSITY

DEPARTMENT OF PHYSICS

Computational Physics (Phys 4203)

Final Exam Solutions

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Problem 1: Area of Grooved Razor Blade (Monte Carlo)

Problem Statement

Determine the area of the shaded region in a grooved thin razor blade shown in Fig. 1. Use **Hit/miss Monte Carlo**.

Fortran Code

```
1  ! =====
2  PROGRAM exam1
3      IMPLICIT NONE
4      INTEGER, PARAMETER :: N=10000
5      REAL :: x, y, A0=9, A, Ac
6      INTEGER :: k, i
7      k=0
8      !let A0 be the area of the square, Ac the area of the half
9      !circles, A the area of the grooved rectangle
10
11      DO i=0, N
12          CALL random_number(x)
13          CALL random_number(y)
14          IF (sqrt(x**2 + y**2) <= 1) THEN
15              k = K + 1
16              !PRINT*, x**2 + y**2, k
17          END IF
18      END DO
19      Ac = 0.5*k/real(N)
20      A = A0 - 4*Ac
21      PRINT*, "The area of the grooved rectangle is:", A, "mm square"
22  END PROGRAM exam1
23  ! =====
```

Listing 1: Fortran code for Hit/Miss Monte Carlo Area Calculation

Results

The calculated area of the razor blade for N=10000 number of random numbers is **A=7.43039989 mm²**. And the number of accepted counts are $N_{acc} = 7848$.

Problem 2: Oxygen Molecule Equilibrium (Lennard-Jones)

Problem Statement

Determine the equilibrium bond distance between atom pairs of the oxygen molecule shown in Fig. 2. Assuming that the primary interaction potential between the atoms is the LJ potential:

$$V = 4\epsilon \left[\left(\frac{\sigma}{d} \right)^{12} - \left(\frac{\sigma}{d} \right)^6 \right]$$

Use $\epsilon = 1.4$, $\sigma = 1.1$. Use molecular dynamics method within employing of **Euler** or **velocity verlet** algorithms.

In case of small oscillation about the equilibrium bond distance, the process may be seen to undergo simple harmonic motion. Plot graphs of U vs t , T vs t and E vs t in such conditions.

Preliminary Mathematical Analysis

To determine the equilibrium bond distance, we start with the Lennard-Jones potential given in the problem:

$$V(d) = 4\epsilon \left[\left(\frac{\sigma}{d} \right)^{12} - \left(\frac{\sigma}{d} \right)^6 \right] \quad (1)$$

The force between the atoms is the negative gradient of the potential with respect to the distance d :

$$\begin{aligned} F(d) &= -\frac{dV}{dd} \\ &= -4\epsilon \frac{d}{dd} (\sigma^{12}d^{-12} - \sigma^6d^{-6}) \\ &= -4\epsilon (-12\sigma^{12}d^{-13} - (-6)\sigma^6d^{-7}) \\ &= \frac{24\epsilon}{d} \left[2 \left(\frac{\sigma}{d} \right)^{12} - \left(\frac{\sigma}{d} \right)^6 \right] \end{aligned}$$

At equilibrium, the net force is zero ($F(d_{eq}) = 0$). Ignoring the trivial solution at $d \rightarrow \infty$:

$$2 \left(\frac{\sigma}{d_{eq}} \right)^{12} - \left(\frac{\sigma}{d_{eq}} \right)^6 = 0 \quad (2)$$

Let $x = (\sigma/d_{eq})^6$. The equation becomes $2x^2 - x = 0$, which implies $x = 1/2$. Thus:

$$\left(\frac{\sigma}{d_{eq}} \right)^6 = \frac{1}{2} \implies d_{eq} = \sigma \cdot 2^{1/6} \quad (3)$$

Substituting the given parameters $\sigma = 1.1$ and $\epsilon = 1.4$:

$$d_{eq} \approx 1.1 \times 1.12246 \approx 1.2347 \quad (4)$$

This theoretical value $d_{eq} \approx 1.235$ will be used to verify the molecular dynamics simulation results.

Small Oscillations (Harmonic Approximation)

For small displacements $x = d - d_{eq}$, the potential can be approximated as a simple harmonic oscillator:

$$V(d) \approx \frac{1}{2}k(d - d_{eq})^2 \quad (5)$$

where the spring constant k is the second derivative of the potential evaluated at equilibrium:

$$k = \left. \frac{d^2V}{dd^2} \right|_{d_{eq}} = \frac{4\epsilon}{\sigma^2} \left(\frac{156}{d_{eq}^{12}} - \frac{42}{d_{eq}^6} \right) = \frac{57.15\epsilon}{\sigma^2} = 72.7 \quad (6)$$

This confirms that for small amplitudes, the motion should exhibit simple harmonic behavior, satisfying $E = T + U = \text{constant}$.

Hence from 5 and 6, we can approximate the force as follows and solve for d :

$$F = -k(d - d_{eq}) = m\ddot{d} \quad (7)$$

Fortran Code

```
1  ! =====
2  MODULE ODE_DRIVER
3      CONTAINS
4
5  SUBROUTINE corrector(t, x, v, dt)
6      IMPLICIT NONE
7      REAL(KIND=16), EXTERNAL :: f
8      REAL(KIND=16), DIMENSION(0:) :: t, x, v
9      REAL(KIND=16) :: dt, k1, k2, k3, p1, p2, p3
10     INTEGER :: i, N
11     N = size(t) - 1
12
13     DO i=1, N - 1
14         t(i) = t(i-1) + dt
15         x(i) = x(i-1) + dt*v(i-1)
16         v(i) = v(i-1) + dt*f(x(i-1))
17
18         k1 = dt*v(i-1)
19         k2 = dt*v(i)
20         p1 = dt*f(x(i-1))
21
22         x(i+1) = x(i-1) + 2*k2
23         p2 = dt*f(x(i))
24         v(i+1) = v(i-1) + 2*p2
25
26         k3 = dt*v(i+1)
27         p3 = dt*f(x(i+1))
28         x(i+1) = x(i-1) + (k1 + 4*k2 + k3)/3
```

```

29         v(i+1) = v(i-1) + (p1 + 4*p2 + p3)/3
30
31     END DO
32
33 END SUBROUTINE corrector
34
35
36 END MODULE ODE_DRIVER
37
38 PROGRAM exam2
39     USE ODE_DRIVER
40     IMPLICIT NONE
41     INTEGER, PARAMETER :: N=10000
42     REAL(KIND=16) :: t(0:N), x(0:N), v(0:N), dt, a=0, c=4e-14
43     REAL(KIND=16) :: E_t(0:N), E_k(0:N), E_p(0:N), xeq=1.235e-10,
44         k=72.7, m=35e-27
45     INTEGER :: p, i
46
47     ! let dx/dt = v, and v0 & x0 the initial conditions
48     ! we also define external function " f " to be used as dv/dt
49     ! = f
50
51     dt = (c-a)/(0.1*N)
52     t(0) = 0
53
54     x(0) = 2e-11
55     v(0) = 0
56
57     CALL corrector(t, x, v, dt)
58
59     E_t = 0.5*m*v**2 + 0.5*k*(x-xeq)**2
60     E_k = 0.5*m*v**2
61     E_p = 0.5*k*(x-xeq)**2
62     OPEN(UNIT=10, FILE="home_exam2.csv", POSITION="APPEND",
63         ACCESS="SEQUENTIAL", STATUS="REPLACE")
64     DO p=0, N-2
65         WRITE(UNIT=10, FMT="(6f32.20)") t(p), x(p), v(p), E_t(p),
66             E_k(p), E_p(p)
67     END DO
68     CLOSE(UNIT=10)
69     PRINT*, x(10), dt
70
71 END PROGRAM exam2
72
73 FUNCTION f(x) RESULT(s)
74     IMPLICIT NONE
75     REAL(KIND=16) :: x, xeq=1.235e-10, s, k=72.7, m=35e-27, r
76     CALL RANDOM_NUMBER(r)
77
78     s = -k*(x - xeq)/m + ((2*r - 1)*1e-7)/m
79 END FUNCTION f

```

76
77

```
=====
```

Listing 2: Fortran code for MD Simulation (Oxygen Molecule)

Results and Graphs

In delivering graphical results for problem 2, I have used the following necessary initial conditions and constants, as can be shown in the above code.

Symbol	Description	Value
m	Mass of an oxygen molecule	$5.31 \times 10^{-26} \text{ kg}$
k	Force constant (calculated)	72.7 N m^{-1}
x_{eq}	Equilibrium position	$1.21 \times 10^{-10} \text{ m}$
x_0	Initial displacement from equilibrium	$2.0 \times 10^{-11} \text{ m}$
t_0	Initial time	0 s

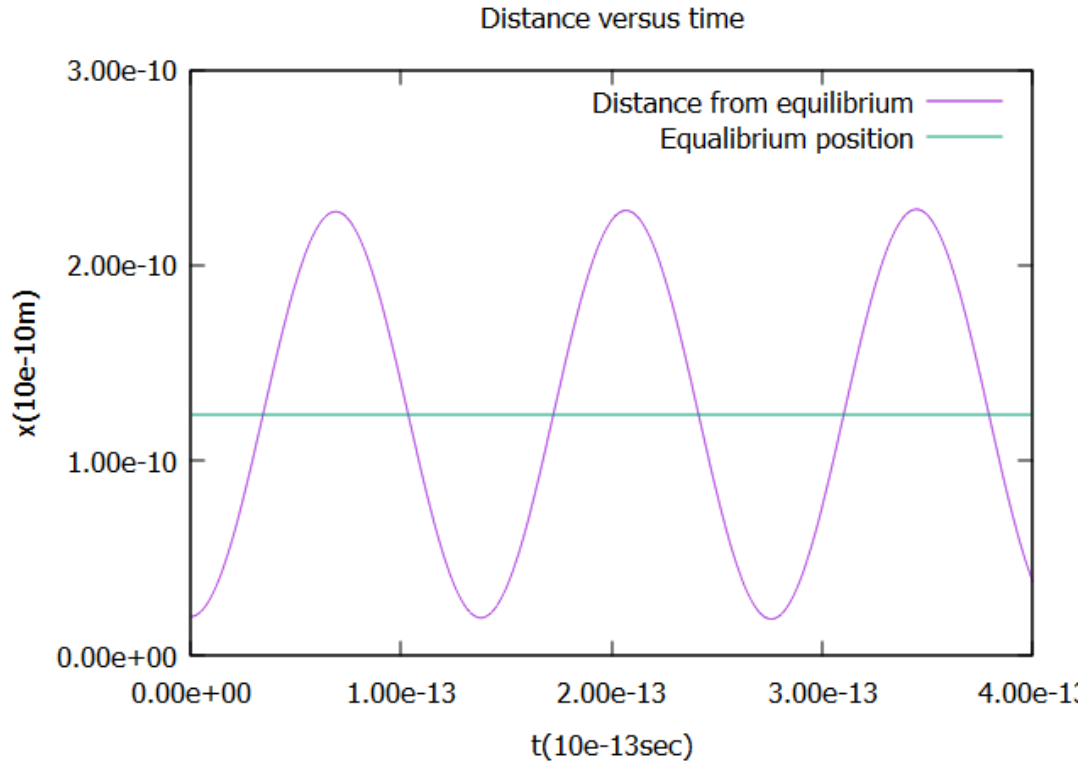


Figure 1: Bond distance d as a function of time.

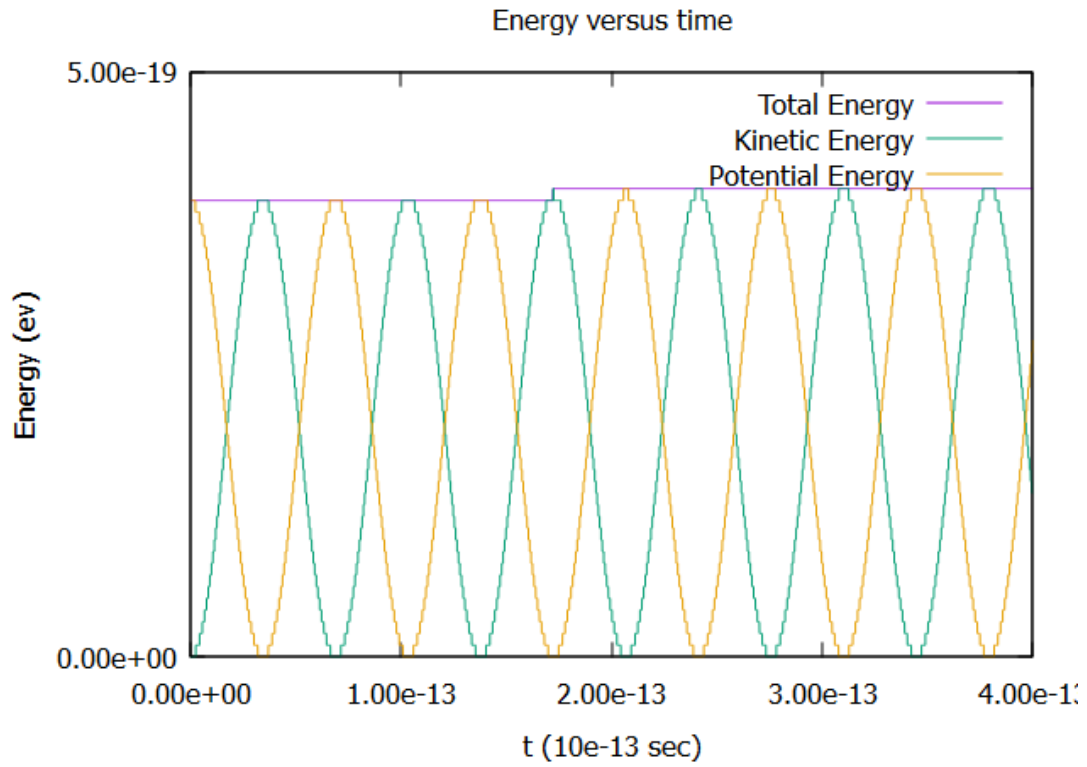


Figure 2: The plot of total energy, potential energy, and kinetic energy as a function of time.

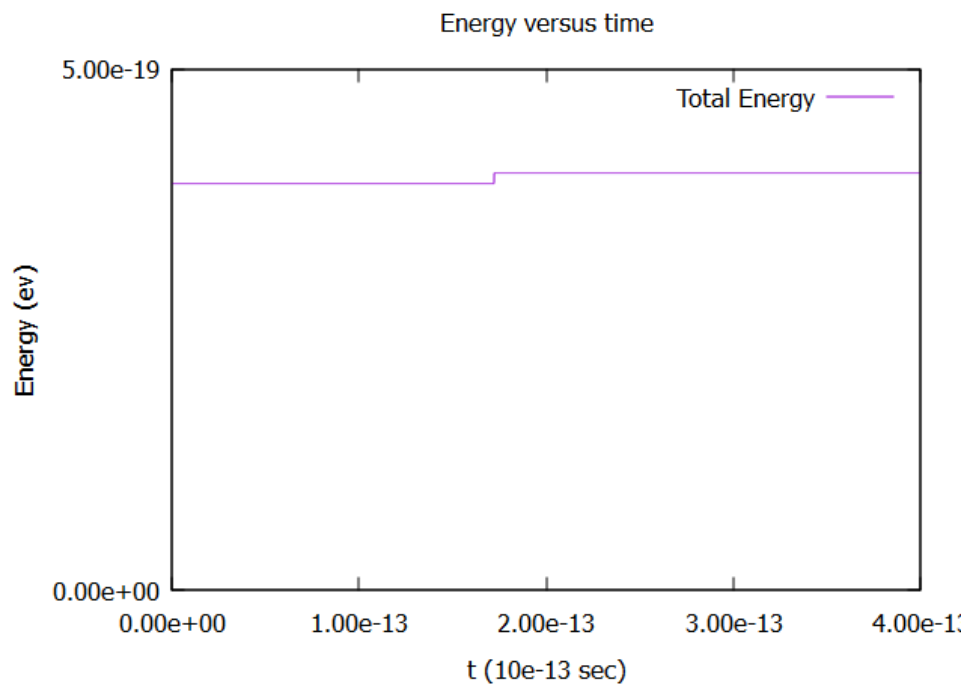


Figure 3: The plot of total energy as a function of time. There is a sudden and slight jump in the energy after a long period of time, perhaps due to a round off errors.

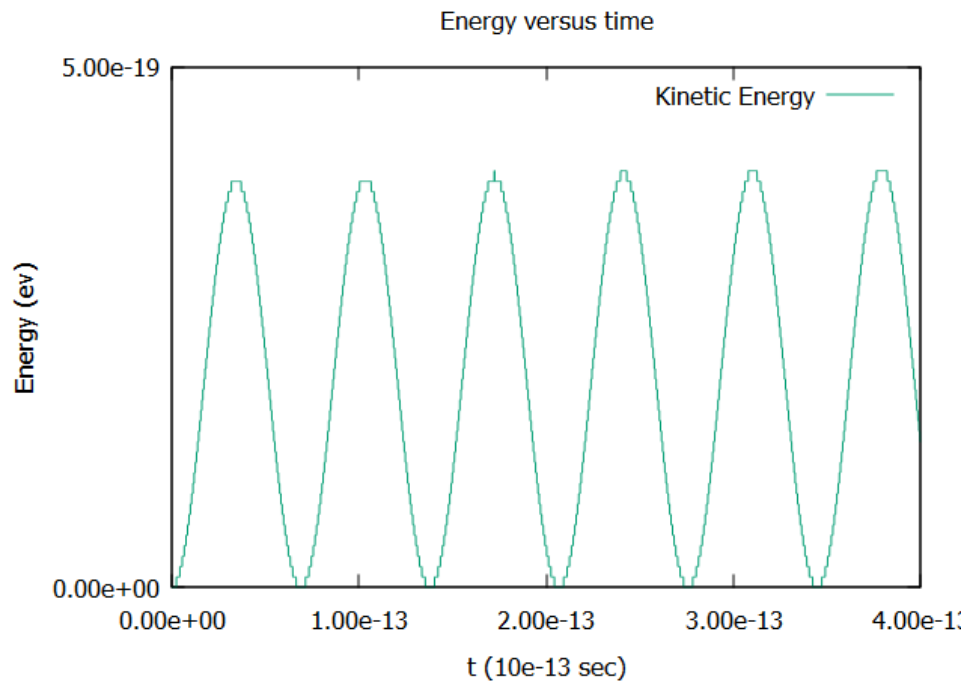


Figure 4: The plot of kinetic energy as a function of time.

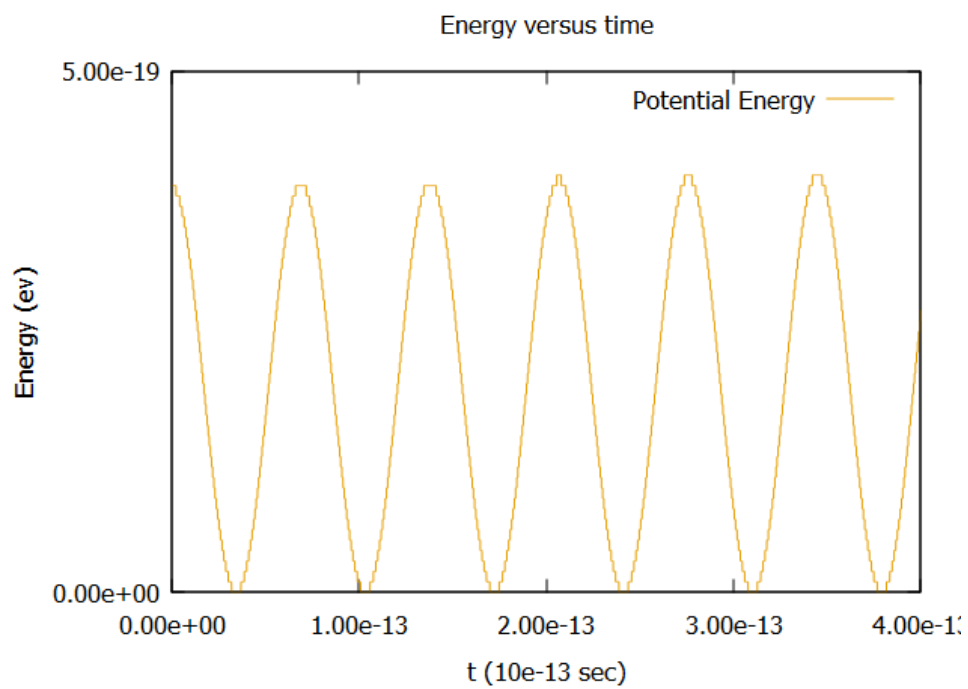


Figure 5: The plot of potential energy as a function of time.

Problem 3: Damped Oscillation with Random Noise

Problem Statement

In problem #2, assume that the oscillation process faces friction resistance with random noise given by:

$$F = -k(d - d_0) + \eta(t), \quad \eta \in (-1, 1)$$

Then show the plot graphs of d vs t , E vs t .

Fortran Code

```
1  ! =====
2  MODULE ODE_DRIVER
3      CONTAINS
4
5  SUBROUTINE corrector(t, x, v, dt)
6      IMPLICIT NONE
7      REAL(KIND=16), EXTERNAL :: f
8      REAL(KIND=16), DIMENSION(0:) :: t, x, v
9      REAL(KIND=16) :: dt, k1, k2, k3, p1, p2, p3
10     INTEGER :: i, N
11     N = size(t) - 1
12
13     DO i=1, N - 1
14         t(i) = t(i-1) + dt
15         x(i) = x(i-1) + dt*v(i-1)
16         v(i) = v(i-1) + dt*f(x(i-1))
17
18         k1 = dt*v(i-1)
19         k2 = dt*v(i)
20         p1 = dt*f(x(i-1))
21
22         x(i+1) = x(i-1) + 2*k2
23         p2 = dt*f(x(i))
24         v(i+1) = v(i-1) + 2*p2
25
26         k3 = dt*v(i+1)
27         p3 = dt*f(x(i+1))
28         x(i+1) = x(i-1) + (k1 + 4*k2 + k3)/3
29         v(i+1) = v(i-1) + (p1 + 4*p2 + p3)/3
30
31     END DO
32
33 END SUBROUTINE corrector
34
35
36 END MODULE ODE_DRIVER
37
38 PROGRAM exam3
39     USE ODE_DRIVER
```

```

40  IMPLICIT NONE
41  INTEGER, PARAMETER :: N=10000
42  REAL(KIND=16) :: t(0:N), x(0:N), v(0:N), dt, a=0, c=4e-14
43  REAL(KIND=16) :: E_t(0:N), E_k(0:N), E_p(0:N), xeq=1.235e-10,
      k=72.7, m=35e-27
44  INTEGER :: p, i
45
46  ! let dx/dt = v, and v0 & x0 the initial conditions
47  ! we also define external function " f " to be used as dv/dt
      = f
48
49  dt = (c-a)/(0.1*N)
50  t(0) = 0
51
52  x(0) = 2e-11
53  v(0) = 0
54
55  CALL corrector(t, x, v, dt)
56
57  E_t = 0.5*m*v**2 + 0.5*k*(x-xeq)**2
58  E_k = 0.5*m*v**2
59  E_p = 0.5*k*(x-xeq)**2
60  OPEN(UNIT=10, FILE="home_exam3.csv", POSITION="APPEND",
      ACCESS="SEQUENTIAL", STATUS="REPLACE")
61  DO p=0, N-2
62      WRITE(UNIT=10, FMT="(6f32.20)") t(p), x(p), v(p), E_t(p),
          E_k(p), E_p(p)
63  END DO
64  CLOSE(UNIT=10)
65  PRINT*, x(10), dt
66
67  END PROGRAM exam3
68
69  FUNCTION f(x) RESULT(s)
70      IMPLICIT NONE
71      REAL(KIND=16) :: x, xeq=1.235e-10, s, k=72.7, m=35e-27, r
72      CALL RANDOM_NUMBER(r)
73
74      s = -k*(x - xeq)/m + ((2*r - 1)*1e-7)/m
75  END FUNCTION f
76
77  =====

```

Listing 3: Fortran code for Damped/Noisy Dynamics

Results and Graphs

Here also I put the constants, I used in my code. All are the same as in porblem 2, but there is an additional constant term " c ". This is added to multiply the noise term, so that the energy scales are comparable with the atomic energy scales.

Symbol	Description	Value
m	Mass of an oxygen molecule	$5.31 \times 10^{-26} \text{ kg}$
k	Force constant (calculated)	72.7 N m^{-1}
x_{eq}	Equilibrium position	$1.21 \times 10^{-10} \text{ m}$
x_0	Initial displacement from equilibrium	$2.0 \times 10^{-11} \text{ m}$
t_0	Initial time	0 s
c	Constant Factor	

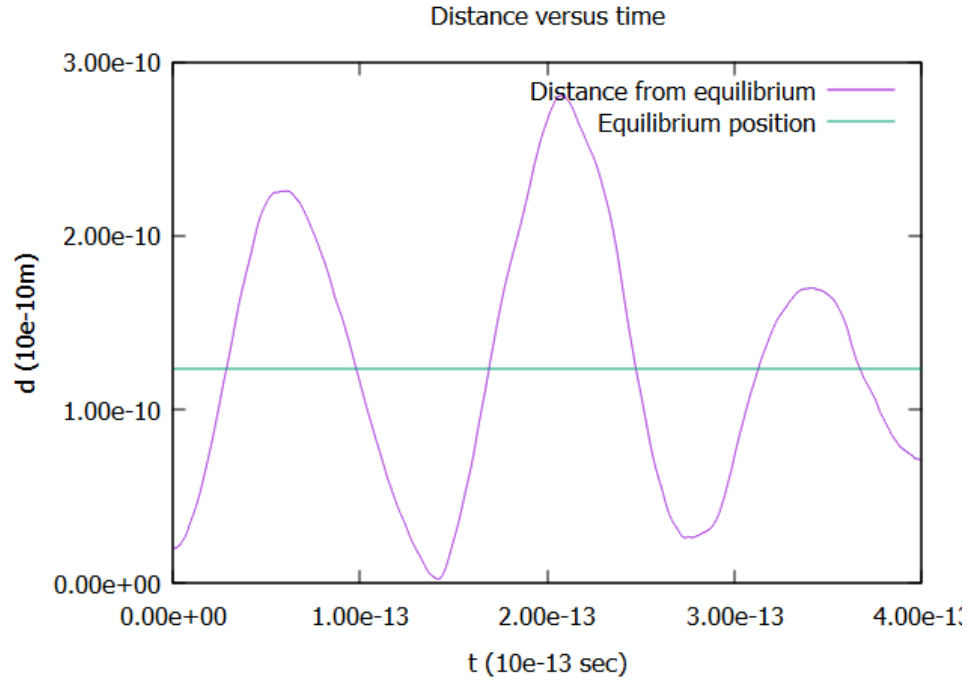


Figure 6: Bond distance vs time under the influence random noise.

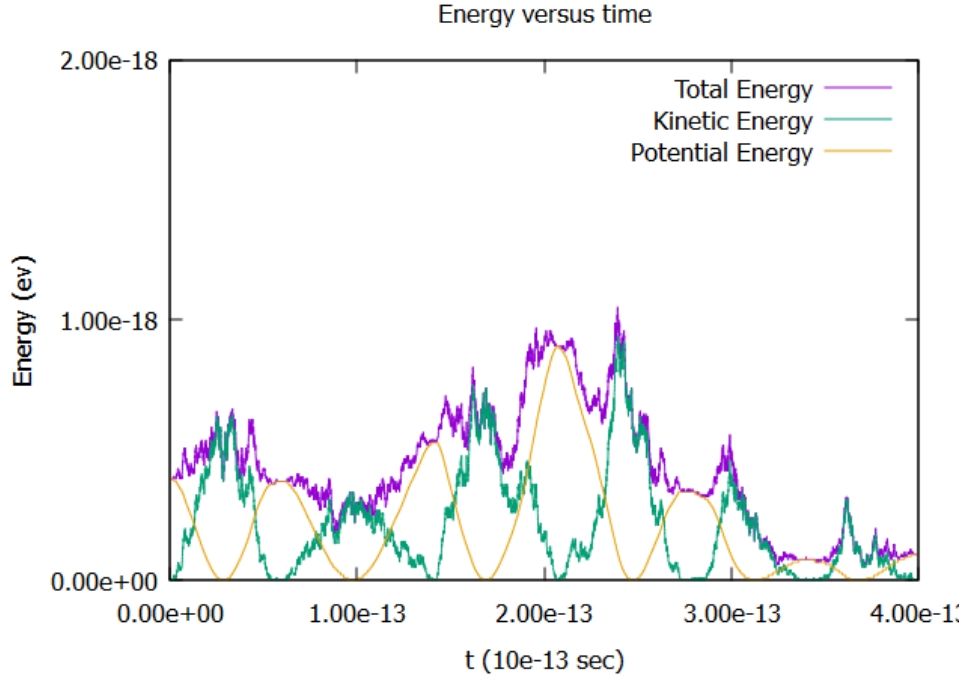


Figure 7: Total Energy, Kinetic Energy, and Potential Energy evolution in the presence and random noise. In sharp contrast to 2 there is a strong fluctuation in the energy of this noise system.

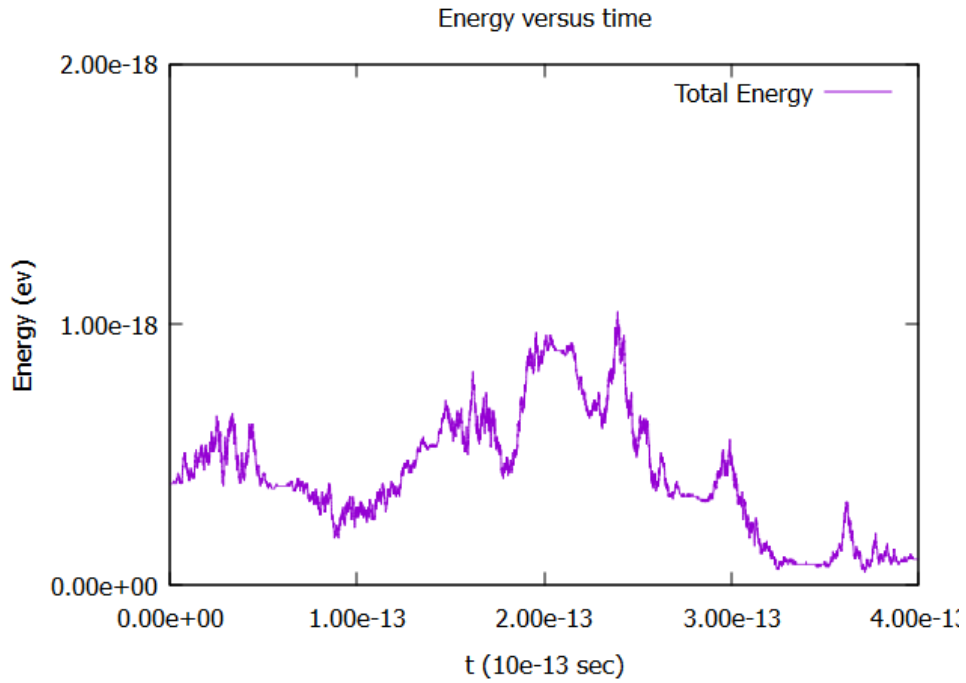


Figure 8: Total Energy evolution in the presence and random noise.